Session 211: Speech and Sensor Signal Processing

SPCOM 2026
International Conference on Signal Processing and Communications

TIME 9:30 AM – 11:00 AM IST VENUE SP 1.20 (TVS Lab)

211.1 **RASE: A Recursive-Adaptive Spectral Enhancer for Multilingual Indian Language Speech Enhancement**
Satya Prasad G, Puja Bharati, Sabyasachi Chandra, Shyamal Das Mandal, Nirmalya Ghosh, Satya Prasad Gaddamedi

211.2 **Tracking Phonetic Transitions in Self-Supervised Speech Representations**
Aastha Kachhi

211.3 **Overcoming NAWEs Fixed-Window Constraints For Spoken Term Detection And Localization**
Ramesh Gundluru, Sankala Sreekanth, Venkatesh Parvathala, Sri Rama Murty Kodukula

211.4 **Semi-Supervised FFT-Spectral Representation Learning for Earthquake Detection in Distributed Fiber Optic Sensing Systems**
Deepika Sasi, Sundaresan Sabapathy, Thomas Joseph

211.5 **Direction of arrival estimation using subspace based iterative orthogonal projection**
Praveen Choppala, Mohammad Sajida Khan

211.6 **Real-Time Correction of Named Entities in Speech Transcriptions: A Lightweight Approach**
Subhadarshi Panda

211.7 **Perceiver encoder for automatic speech recognition**
Kaustubh Kulkarni, Venkateswarlu Gonuguntla

211.8 **Phoneme-Preserving Mamba-Based Frontend for Noise-Robust Automatic Speech Recognition**
Reshma K R, Anil Vuppala

211.9 **Spoken Language Diarization using Source-Domain Aggregative Weakly-Supervised Language Labels**
Spandan Dey, Hirak Mondal, Premjeet Singh

211.10 **Exploiting Compressed Eigencomponent Transformation in DNN Framework for Passive SONAR Self-noise Cancellation**
Pawan Kumar, Karan Nathwani

211.11 **Impact of Localization Errors on Deep Learning Models for Wideband Spectrum Monitoring**
Aman Kumar, Jay Vikrant, Sundaram Vanka, Sumohana Channappayya

211.12 **QMOS: Qwen-based MOS Prediction for TTS**
Ravi Sastry Kolluru, Sri Charan Devarakonda, Radhe Shyam Salopanthula, Anil Vuppala

211.13 **Validating Smartphone-Based 3D Head Models for Precision EEG Source Localization**
Suranjita Ganguly, Malaaika Chhaya, Mohan Raghavan

211.14 **Audio-Based Urgency Detection with SHAP-Enhanced Feature Refinement**
Ankit Dighe, Sanika Gadkari

211.15 **Leveraging Deep Bayesian Inference for Efficient Multiple Source Localization with Spherical Harmonics**
Priyadarshini Dwivedi, Gyanajyoti Routray, Rajesh M Hegde

Session 212: Secure and Terahertz Communication Systems

SPCOM 2026

International Conference on Signal Processing and Communications

TIME 9:30 AM – 11:00 AM IST VENUE SP Building Side Foyer

212.1 **Variational Bayes Estimation for Affine-Precoded Superimposed Pilots in Partially Connected Dual-Wideband Tera-Hertz MU-MIMO Systems**

Abhisha Garg, Suraj Srivastava, Aditya Jagannatham

212.2 **Improving the Cheater Identification Capability of Threshold Secret Sharing Schemes**

Tanmoy Mandal, Srinivasan Krishnaswamy, Ratnajit Bhattacharjee

212.3 **Finite-Key Analysis of Phase-Matching QKD Type Protocol over a Satellite Channel under Atmospheric Turbulence**

Pranav Surapuraju, Lokendra Chouhan

212.4 **End-to-End Post-Processing for CV-QKD with Heterodyne Detection: ADC Modelling, Amplitude Sifting, Tri-Mode NB-LDPC Decoding, and Selective Privacy Amplification**

Mohammad Sikander sheikh, Rohit Chaurasiya

212.5 **EM-based Model for RIS-aided Channel and its Achievable DoF**

Nobhendu Chowdhury, Praful Mankar

212.6 **Regularized Multi-Measurement Sparse Decomposition for Time-Domain Ambisonics Upscaling**

Gyanajyoti Routray, Priyadarshini Dwivedi, Rajesh M. Hegde

212.7 **Terahertz Topological Valley Photonic Crystals with Polymer-Filled Bearded Interface for Robust Waveguiding in 6G Applications**

Dibaskar Biswas, Tushar Shah, Sourodipto Das, Amit Baran Dey, Ashish Kumar Chowdhary, Debabrata Sikdar

212.8 **Topologically Protected Supercoupling in Cavity-Integrated Photonic Structure for Terahertz Communication**

Tushar Shah, Dibaskar Biswas, Lokesh Sharma, Sourodipto Das, Ashish Kumar Chowdhary, Debabrata Sikdar

Session 213: Signal and Image Processing

TIME 9:30 AM – 11:00 AM IST VENUE ECE 1.07

SPCOM 2026
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213.1 **Convolution Based Fast Lossless Compression Algorithm for Color Filter Arrays**
Chiranjeev Bhaya, Manish Kumar, Chandra Mohan Velpula, Praveen Bangre Prabhakar Rao

213.2 **Modelling Bandlimited Signals in Self-Dual Differential Equations**
Abhijat Bharadwaj, Animesh Kumar, Shaileshram Sivakumar, Advay Bapat, Vikram Gadre, Shrikant Sharma, P. Radhakrishna, Ram Bilas Pachori

213.3 **Hierarchical Convolution Spatial Propagation Networks for Depth Completion**
Praveen Ramesh, Vikas Gowda, Kunal A Kathare, Basavaraja Shanthappa Vandrotti, Jooyoung Kim, Ankit Dhiman

213.4 **Bilevel Learning of Task-Adapted Regularizers**
Aryan Saha, Subhadip Mukherjee

213.5 **Consecutive Support Matching Induced Parameter Tuning Accelerates Momentum Iterative Hard Thresholding**
Samrat Mukhopadhyay, Debasmita Mukherjee

213.6 **Physics-Guided Residual-Aware Consensus Equilibrium for SAR Image Despeckling**
Satyakam Baraha, Abhijit Mishra, Ajit Kumar Sahoo

213.7 **Consistency is Key to Image Denoising**
Oindrila Haldar, Saptarshi Mandal, Chandra Sekhar Seelamantula

213.8 **Accelerating Quaternion Low-Rank Approximation for Color Images via Complex Embedding**
Akshadha A, Devika Rani J, Sooraj K. Ambat

213.9 **Hard Thresholding Based LMS Algorithm for Adaptive Graph Signal Recovery**
Rachita Mohanty, Rajib Lochan Das

213.10 **Mixing Time Bounds for Asymmetric Random Walks on Hypercubes, Cycles, and Tori Using Coupling**
Mrudula Mahindrakar, Hrushikesh Kant, Avhishek Chatterjee

213.11 **A Small-World Network Based Framework for Texture Classification**
Shubham Dwivedi, Keerthika Kadagala, Om Jee Pandey, Rajesh M. Hegde

213.12 **Power Series Based Iterative Algorithms for Distributed Sparse Recovery**
Shashank S, Ketan Atul Bapat, Mrityunjoy Chakraborty